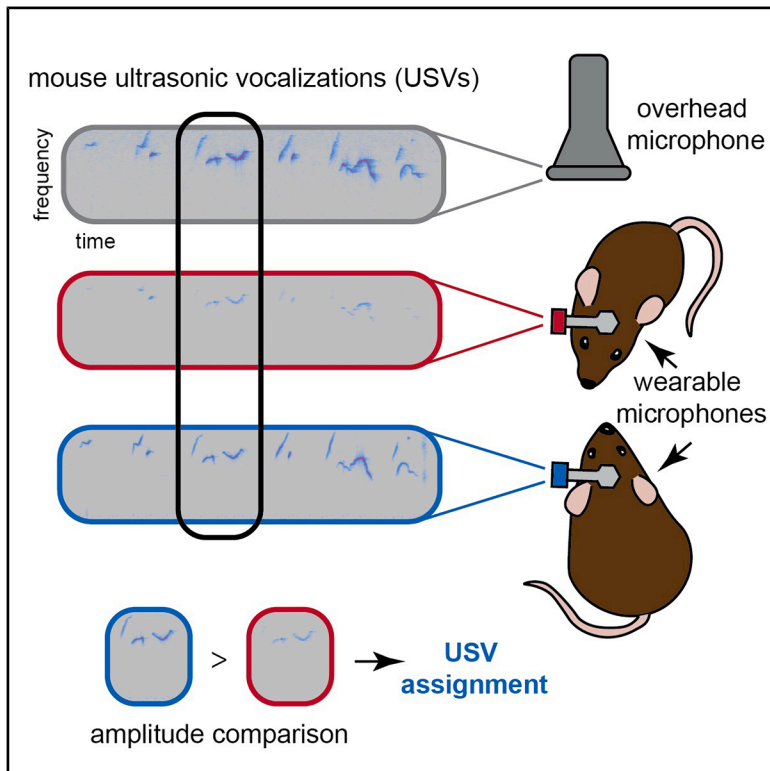


# Mountable miniature microphones to identify and assign mouse ultrasonic vocalizations

## Graphical abstract



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## In brief

Mouse ultrasonic vocalizations (USVs) are used to study vocal behavior. Here, Waidmann et al. develop a wearable miniature microphone system to assign each USV to a specific mouse in a pair. Using mini-microphones with integrated video tracking, the authors demonstrate that female mice rarely produce USVs in a male-female pair.

## Highlights

- Mountable miniature microphones allow assignment of mouse ultrasonic vocalizations
- System assigns 89% of mouse USVs with only 2 microphones and 97% with video correction
- In male-female pairs, males produce >96% of USVs, while female USVs are rare (<1%)



## Article

# Mountable miniature microphones to identify and assign mouse ultrasonic vocalizations

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**MOTIVATION** Mouse ultrasonic vocalizations (USVs) have been a useful mammalian model for studying vocal production and social behavior. However, it is difficult to assign vocalizing identity in pairs of animals without the use of complex microphone arrays or the assumption that only a single animal is vocalizing. We present a wearable miniature microphone system that can reliably detect and assign vocalizations to individuals in close proximity with high accuracy. This allows us to examine the behavioral dynamics of mouse courtship behavior with greater detail and demonstrate the system's potential in studying vocal behavior across a wide variety of species.

## SUMMARY

Mouse ultrasonic vocalizations (USVs) are a promising model for studying vocal production. Although, in courtship interactions, males emit the majority of the USVs, female mice also produce USVs. In order to study the mechanisms of vocal production in freely behaving mice, it is necessary to identify the individual responsible for each syllable. Prior studies have used microphone arrays, but these are costly and require complex analyses to pinpoint the vocalizer. Here, we developed an inexpensive, ultrasound-sensitive wearable microphone to identify USVs from individual mice in socializing pairs. We reliably detected USVs and assigned 90% to a specific animal in a pair based on relative amplitude differences. When paired with video tracking, we increased the assigned percentage (97%) and described the courtship behavioral landscape in which USVs occur. These results offer a low-cost, simple method to study social communication between pairs of mice and other ultrasonically vocalizing animals.

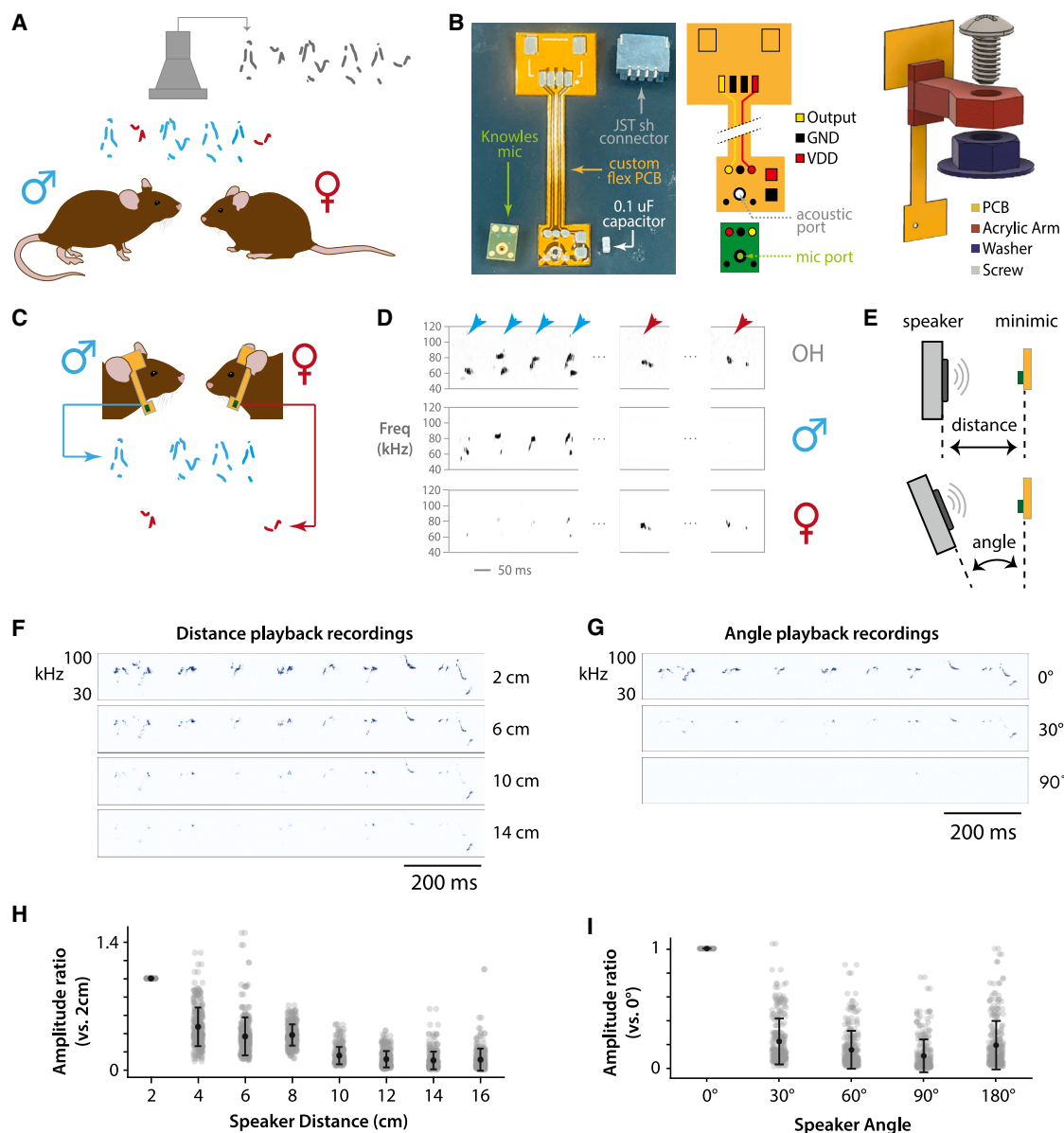
## INTRODUCTION

Vocalization is a widespread form of communication across the animal kingdom, used to convey information about predators, reproductive fitness, and courtship readiness. While many species produce low-frequency audible vocalizations, including songbirds and humans, many also produce vocalizations in the ultrasonic range across a variety of social contexts.<sup>1–14</sup> The house mouse, *Mus musculus*, produces both low-frequency squeaks in response to aversive stimuli and social ultrasonic vocalizations (USVs) to communicate with conspecifics.<sup>1,9,15</sup> USVs are composed of discrete syllables, formed into longer “song” bouts, and are mostly innate,<sup>16,17</sup> where individual syllables are first produced from birth by pups to initiate maternal retrieval when separated from the nest.<sup>18,19</sup> After around postnatal day 12, when mice rarely produce isolation USVs, young adults begin to produce social USV sequences.<sup>20,21</sup> In particular, male mice

readily vocalize for long bouts during female-directed courtship,<sup>9</sup> and these vocalizations drive female preference and courtship behavior.<sup>22–25</sup> Males do demonstrate some flexibility in their vocal usage and repertoire depending on social context<sup>24,26,27</sup> and signatures of individual identity across different mice,<sup>28</sup> which suggests that USV production is a dynamic and complex social behavior. Changes to USV production are also frequently used to measure social deficits in mouse models of human neuropsychiatric conditions,<sup>29–34</sup> making them a potentially broader model system for mammalian vocal and social behavior. Studies of the mechanisms underlying USV production provide insight into neural and genetic drivers of innate vocal production behavior.<sup>35–41</sup> However, to fully study the mechanisms of production of social USVs, a full picture of the social landscape, including all individual vocal contributions, is required.

While much of the study of mouse ultrasonic vocal behavior is centered on male courtship songs, females are also capable of





**Figure 1. Wearable mini-microphone design and recording in social pairs**

(A) Identity assignment problem during multi-animal mouse ultrasonic vocalization (USV) sessions. A single ultrasonic overhead (OH) microphone used to record USVs in a social context (gray) cannot discriminate between syllables that originate from the male (blue) or the female (red).

(B) Design and layout of the mini-microphone recording device. Left, photograph of the mini-microphone circuit board and JST connector against a blue paper background. Right, 3D model of the custom attachment arm, consisting of the acrylic arm (red) secured between a washer (blue) on the mouse's skull and a screw (gray).

(C) Schematic of male and female mice with attached mini-microphones, each able to parse out vocalizations from the individual mouse. The printed circuit board (yellow) is glued to the acrylic arm, with the mini-microphone acoustic port pointed toward the mouse's mouth.

(D) Segments of spectrograms from mini-microphone recordings, showing vocalization assignment. Top, recording from OH microphone collecting all vocalizations. Middle, recording from mini-microphone attached to the male mouse. Bottom, recording from mini-microphone attached to the female mouse. Amplitude differences between male and female spectrograms were used to assign individual vocalizations as male (blue arrows) or female (red arrows).

(E) Schematic of the mini-microphone distance and angle tests. An audio file of recorded USVs was played on an ultrasonic speaker (gray) and recorded on the mini-microphone (yellow) at varying angles and distances between the speaker and microphone.

(F) Spectrograms at different recording distances, between 2 and 16 cm at 2-cm increments. A 1-s excerpt for the recording is shown. Amplitude visibly drops off with distance.

(G) Spectrogram segments taken at different recording angles (0°–180° at 30° increments) at a distance of 6 cm. The same recording is used as in (F). Amplitude visibly drops off with greater angle deviation.

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producing USVs, both in male-female<sup>42–45</sup> and female-female interactions.<sup>46</sup> Prior research in USV production behavior has focused mostly on male vocalizations, both in their production of USVs and in their behavioral response to external USVs. In order to study the complex behavioral interactions between two or more vocalizing animals, it is necessary to parse which vocal syllables are produced by which individual, which is impossible with a traditional recording setup using a single overhead microphone (Figure 1A). USVs are produced at close range<sup>43</sup> and at high frequency.<sup>47</sup> Recently, to overcome this USV assignment problem, several labs have developed array systems of microphones surrounding or above a recording arena.<sup>42–45</sup> Complex mathematical and computational tools are then used to sort out which vocalizations belong to whom. While these methods increase animal identification fidelity, it is expensive to require multiple high-fidelity microphones in a single recording arena and difficult to computationally and analytically triangulate the source of individual USVs, especially those produced when animals are in close proximity. In other vocalizing species, particularly songbirds<sup>48</sup> and bats,<sup>49</sup> researchers have developed wearable microphones that individuals carry and whose amplitude differences can be used to assign the identity of a single vocalization. However, these existing technologies are designed to detect low-frequency vocalizations and do not capture mouse vocalizations in the ultrasonic range (30–100 kHz). We sought to develop similar wearable microphones for mice in order to simplify the assignment problem for mouse USVs.

Here, we identified an ultrasound-capable miniature microphone and developed custom hardware and USV assignment analysis. Overall, the system is very low cost and easy to implement. It is also customizable and can be integrated into existing recording systems. We used this system to record extended courtship sessions in multiple pairs of freely behaving mice and demonstrate that our system can, with a very simple analysis, assign a high percentage of USVs to individual animals. We confirm that females do produce USVs in extended social interactions with males but at much lower rates. Finally, we integrate the vocal behavior with video data to further increase our assignment accuracy and identify differences in animal positions during the production of male and female vocal interactions to build a larger picture of mouse social behavior.

## RESULTS

### Wearable microphones can measure mouse USVs

We attempted to solve the USV assignment problem in a high-fidelity manner while minimizing cost and computational investment. We identified a commercially available small (2.75 [length] × 1.85 [width] × 1.00 [height] mm) and lightweight (0.03 g) micro-electrical mechanical system (MEMS) micro-

phone, normally used for detecting high-frequency sounds to diagnose problems in electronics or buildings (Knowles SPU0410LR5H-QB; Figure 1B).

We first found that the mini-microphone was capable of detecting mouse USVs played from an ultrasonic speaker (Avisoft BioAcoustics Ultrasonic Dynamic Speaker with 216H US playback interface) at the same decibel (loudness) level that mice normally produce USVs (Figure 1E). Importantly, when the distance between the mini-microphone and the speaker increased, the recorded amplitude decreased, and the detection level dissipated at a distance of about 10 cm from the speaker (Figures 1F and 1H). Further, when the angle of the mini-microphone was changed from directly parallel to the speaker to 30°–180° degrees to the speaker, the playback vocalizations were much less detectable (Figures 1G and 1I; more details are provided in Figure S1). These findings suggest that the ultrasonic mini-microphones are very sensitive to the angle and distance from which the sounds originate.

We next attached the mini-microphone to a custom printed circuit board (PCB), which in turn we attached to a 3D-printed mounted headgear; the audio signal was transferred through a micro-JST connector to an Ultrasound Gate 416H (Avisoft) recording system (Figure 1B). The total weight of the microphone, circuit, and headgear mount is 1.17 g, rendering it easily wearable. We then mounted the mic headgear to a screw mount on the mouse skull under the skin (placed and glued earlier during surgery). For each individual mouse, the acoustic port was close to 0° to the mouth (Figures 1C and S1H) and thus 90°–180° to another animal if face to face.

### USV assignment with miniature microphones in male-female dyads

We recorded 9 naturalistic, free-behavior courtship sessions, each between one male and one female mouse, each with a mounted mini-microphone, ranging from 20 to 40 min in duration. We used 4 males and 2 females, with 3 pairs recorded twice. In this setup, we were able to record male and female USVs and observe amplitude differences between the mini-microphones (Figure 1D). We collected 22,561 USV syllables in total (a full list of individual pairings and USVs per pairing is in Table 1).

We also recorded concurrent video data in order to align the recorded vocal behavior with broader social and courtship behavioral motifs. To perform these video recordings, we designed and constructed a recording box from opaque white acrylic with overhead openings for an ultrasonic microphone, overhead camera, and mini-microphone cables (Figure 2A). The box had a clear acrylic base and was suspended on an external aluminum frame to allow a recording from an additional bottom-view camera (see method details). We used a manual cable rotator to avoid tangling between the mini-microphone cables of the two freely moving animals.

(H) Comparison of each syllable amplitude between 2 cm and each other distance. For each individual syllable, at each distance, syllable amplitude was averaged within the time range identified on the 2-cm recording. For every syllable, the amplitude at distance  $d$  is divided by the amplitude of that same syllable at the 2-cm distance (ratios are all 1 at 2 cm). Black line, mean  $\pm$  1 standard deviation.

(I) Comparison of each syllable amplitude between 0° and each other angle. Analysis is conducted using the same procedure as the angle in (H) but relative to the 0° distance.

**Table 1. Distribution of USV assignments in male and female pairs using the mini-microphones**

| Session (M_F) | Total USVs | Male USVs (%) | Female USVs (%) | Unass. USVs (%) | Total USVs (during video) | Male USVs (distance corrected) (%) | Female USVs (distance corrected) (%) | Unass. USVs (distance corrected) (%) |
|---------------|------------|---------------|-----------------|-----------------|---------------------------|------------------------------------|--------------------------------------|--------------------------------------|
| B770_B744 (1) | 6,691      | 5,821 (87.0)  | 178 (2.7)       | 692 (10.3)      | 5,674                     | 5,491 (96.77)                      | 21 (0.37)                            | 162 (2.86)                           |
| B757_B744     | 621        | 553 (89.0)    | 32 (5.2)        | 36 (5.8)        | 587                       | 566 (96.42)                        | 9 (1.5)                              | 12 (2.04)                            |
| B770_B745     | 3,688      | 3,283 (89.0)  | 77 (2.1)        | 328 (8.9)       | 3,321                     | 3,215 (96.81)                      | 7 (0.21)                             | 99 (2.98)                            |
| B782_B744 (1) | 2,575      | 2,420 (94.0)  | 26 (1.0)        | 129 (5.0)       | 2,517                     | 2,351 (93.40)                      | 17 (0.68)                            | 149 (5.92)                           |
| B782_B744 (2) | 1,248      | 690 (55.3)    | 303 (24.3)      | 255 (20.4)      | 1,114                     | 1,031 (92.55)                      | 40 (3.59)                            | 43 (3.86)                            |
| B782_B745 (2) | 2,740      | 2,487 (90.8)  | 76 (2.8)        | 177 (6.5)       | 1,043                     | 1,029 (98.66)                      | 5 (0.48)                             | 9 (0.86)                             |
| B782_B745 (1) | 1,983      | 1,593 (80.3)  | 135 (6.8)       | 255 (12.9)      | N/A                       | N/A                                | N/A                                  | N/A                                  |
| B770_B744 (2) | 857        | 551 (64.3)    | 40 (4.7)        | 266 (31.0)      | N/A                       | N/A                                | N/A                                  | N/A                                  |
| B805_B745     | 2,158      | 1,443 (66.9)  | 389 (18.0)      | 326 (15.1)      | N/A                       | N/A                                | N/A                                  | N/A                                  |
| Total         | 22,561     | 18,841 (83.6) | 1,256 (5.6)     | 2,464 (10.9)    | 14,256                    | 13,683 (95.98)                     | 99 (0.69)                            | 474 (3.32)                           |

Unass., unassigned.

With the overhead microphone and respective mini-microphones, a total of 3 sound channels were recorded during each session and converted to spectrograms (male, female, and overhead). We used the overhead microphone as a ground-truth record for all emitted vocalizations. To compute syllable amplitude, for each spectrogram time point, we identified the 3 frequencies with the highest amplitude on the overhead channel to avoid capturing any mechanical or electrical noise that may have been present. We computed the mean amplitude on the male and female channels across those frequencies for each time point (Figure 2B). We performed syllable identification and identified the start and end of each USV syllable on the recording from the overhead microphone using Mouse Song Analyzer 2 (MSA2).<sup>50</sup> By measuring the median amplitude on the separate male and female recording channels across the duration of syllables, we obtained a median male syllable amplitude ( $M_{amp}$ ) and median female syllable amplitude ( $F_{amp}$ ) for each USV. Finally, we took the log of the ratio between male and female syllable amplitudes,  $\log(M_{amp}/F_{amp})$ , to generate an amplitude ratio metric for each syllable (see method details).

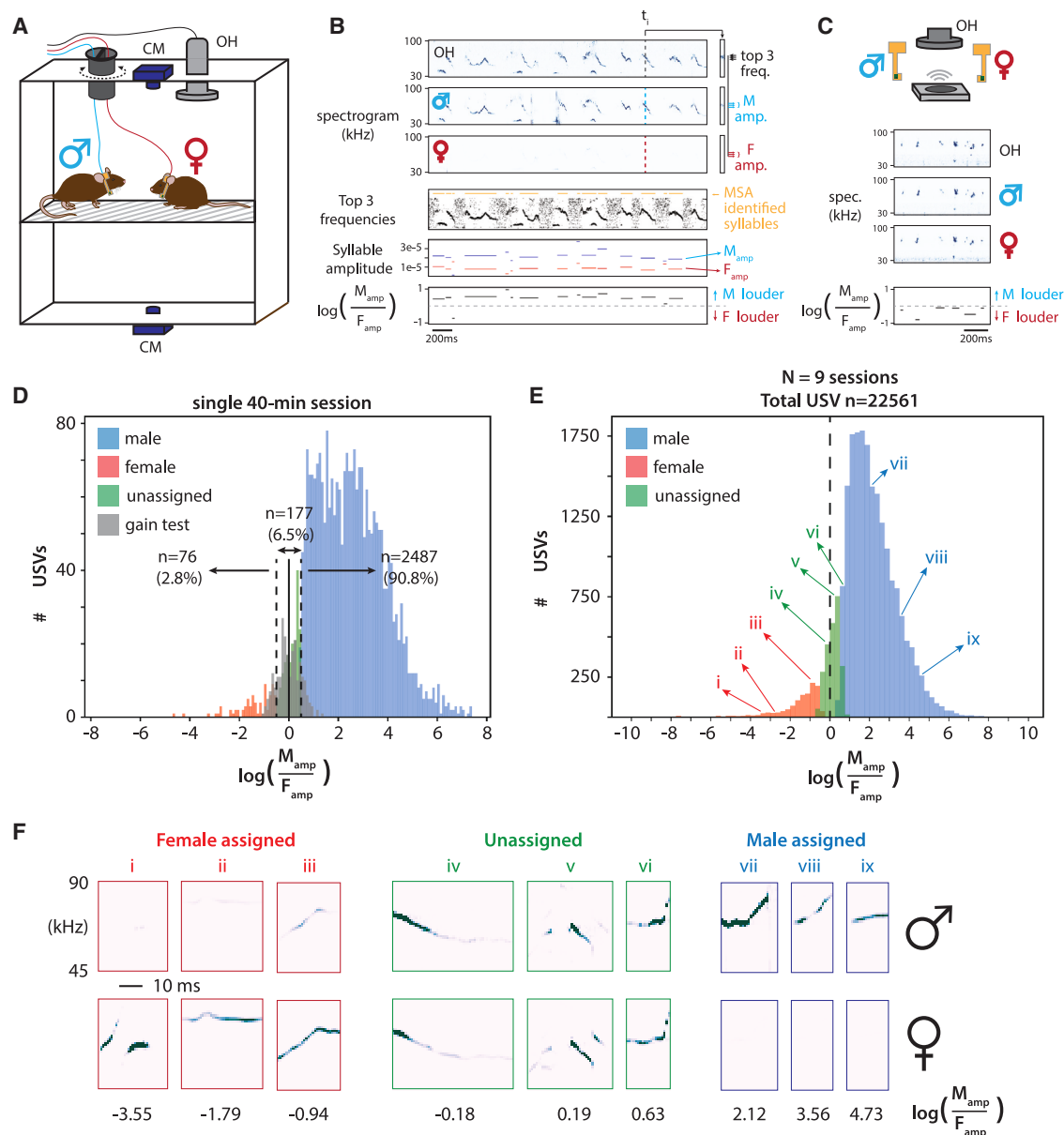
Because the sensitivity of individual mini-microphones may vary across sessions, the amplitude ratio alone was not sufficient to assign all USVs as either male or female emitted. To compensate for these inherent variations, we ran a brief microphone calibration procedure with USVs playing from the ultrasonic speaker, equidistant to each different microphone, prior to each recording session (Figure 2C). Gains of both mini-microphones were manually adjusted such that a playback recording of mouse USVs had visually similar amplitudes on both channels. We recorded 30–60 s of the playback on all three recording channels before each male-female recording session. We then computed the male-female amplitude ratio (as above, Figure 2C) for each syllable recorded in the gain test session. We used the distribution of these amplitude ratios, gain matched and from a single audio source, as a baseline against which to measure male and female vocalizations produced during the social recording. If a syllable in the social recording had a  $\log(M_{amp}/F_{amp})$  greater than 1 standard deviation above the

mean amplitude ratio from the gain test, we assigned it as male; if  $\log(M_{amp}/F_{amp})$  was less than 1 standard deviation below the gain test mean, we assigned it as female; and if it fell within 1 standard deviation of the gain test mean, we designated it as unassigned.

An example of this process shows the distribution of amplitude ratios for a single 40-min male-female session, in which 90.8% of vocalizations were assigned as male emitted, 2.8% as female emitted, and the remaining 6.5% as unassigned (Figure 2D). Across all 9 male-female sessions of 22,561 unique vocalizations, 83.6% (range: 55.3%–94.0%) were assigned as male emitted, 5.6% (range 1.0–24.3) as female emitted, and 10.9% (range 5.0–31.0) as unassigned (Figures 2E and S2; Table 1). Figure 2F shows spectrograms of example male, female, and unassigned syllables from across the range of amplitude ratios.

We next performed a control experiment, in which we placed microphones on both male and female mice as above, but then during the session ( $n = 2$ ), we anesthetized the female mouse before placing the male in the setup; in this condition, all vocalizations should be male assigned. However, we still detected some female-assigned USVs, primarily when the male was at a close inter-animal distance to the female (Figure 3B). This meant that some of the vocalizations from one animal were being picked up as louder by the mic from the other, even after calibration. We sought to use our integrated video data to perform a distance correction because, in playback recordings, the miniature microphones showed sensitivity to distance (Figure 1). For 6 of the male-female courtship sessions, we recorded and aligned the video of the mice with the audio output. We trained an SLEAP animal pose tracking model<sup>51</sup> to track various body parts on each mouse (Figure 3A). We computed the inter-animal distance during each USV ( $d_{USV}$ ) between animal centroids and then normalized them to maximum and minimum distance ( $d_{max}$  and  $d_{min}$ ). We then applied a distance-correction scalar to our amplitude ratio as follows:

$$\log\left(\left(M_{amp}/F_{amp}\right) * \left(d_{max} - d_{min}/d_{USV} - d_{min}\right)^2\right).$$



**Figure 2. Behavioral setup and vocalization assignments**

(A) Schematic of custom behavioral rig for paired social recordings (male ♂, blue; female ♀, red). Each mouse wears a mini-microphone (yellow) and is allowed to freely interact in an acrylic box (white) on a clear acrylic floor (gray dashes). An overhead ultrasonic microphone (gray, "OH") records all vocalizations that occur, while the top and bottom cameras (blue, "CM") record movement. Mini-microphone cables (red, blue) are run through a manual cable untangler (black).

(B) Computing male-female amplitude differences. Top 3 rows, spectrograms of recordings from the overhead, male mini-microphone, and female mini-microphone, respectively. 4th row, the top 3 frequencies from the overhead recording, with start and end times of each syllable identified using MSA 2.0 (orange). 5th row, median male mini-microphone amplitude (blue) and female mini-microphone amplitude (red) averaged across the time and frequency segments shown in row 4. 6th row,  $\log$  of male:female amplitude ratio between two individuals.

(C) Pre-session gain calibration. Top, schematic of speaker playing USVs to overhead, male, and female microphones. Spectrograms demonstrate signals of roughly equivalent amplitude across all three recording channels. As in (B), and a male mic:female mic amplitude ratio is calculated for each syllable.

(D) Histogram displaying the male:female amplitude ratios of all syllables across a single 40-min recording session. The distribution of the ratios from the gain-matching session (gray, solid red line: mean;  $\pm 1$  standard deviation: dashed red line) was used as the point of comparison for amplitude differences in the courtship session. Male-assigned (blue), female-assigned (red), and unassigned (green) syllables based on difference from gain-matching session mean.

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After applying this distance correction, when the female was anesthetized, we attributed 0% of the USVs to the female and 99.6% to the male (Figure 3B). When we applied this distance correction to the original 6 sessions, for 14,256 USVs that occurred during video recording, we attributed 95.98% (range: 92.55%–98.66%) of USVs to the male, 0.69% (range: 0.21%–3.59%) to the female, and 3.32% (0.86%–5.92%) as unassigned (Figure 3B; Table 1). The resulting distribution indicates a further shift in assignment toward males compared to our initial analysis (Figure 3C). Adding a video component and distance correction thus not only dramatically improves assignment but also suggests that many USVs produced at close range in male-female pairs, while attributed to the female,<sup>42–45</sup> may in fact be produced by the male.

### Sex-specific behavior during USV production

We found that USVs were produced primarily when animals were in very close proximity (most often 0–10 cm apart; Figure S3C) and occurred in repeated bouts throughout the length of the 40-min sessions (Figure 4A). By using our USV assignment system, paired with our distance correction, we identified the individual producing each vocalization and could therefore calculate the inter-animal distance during male, female, or unassigned USVs.

We found that while males most often produced USVs when they were very close to the female (less than 10 cm), they still produced USVs within the entire distance range (some even at nearly 35 cm apart). Females also produced USVs across the full range in the arena, but in contrast to males, they mainly produced USVs at much greater distances from males (20–40 cm; see example in Figure 4A, bottom). These distance profiles and relationships were significant and were across all 6 pairs; the mice were significantly closer during USV vs. non-USV production frames (independent t test; vocalizing vs. random nonvocalizing frames  $p = 0$ , vocalizing vs. all nonvocalizing frames  $p = 0$ , and random nonvocalizing vs. all nonvocalizing frames  $p = 0.3$ ), and female (F) and unassigned (UN) vocalizations were produced at significantly farther distances than male (M) vocalizations (independent t test; M vs. F  $p = 0$ , M vs. UN  $p = 0$ , and F vs. UN  $p = 0.023$ ; Figures 4B and S4). To better define body positions during USV production behavior, we also measured the distance from the male nose to the female nose and to the female tail base during USV frames. We found that male vocalizations most commonly occurred when their nose was in extremely close proximity to the female ano-genital region, with a smaller proportion produced when the animals were nose to nose (Figure 4C). These findings suggest that males are more motivated to produce courtship USVs when chasing females with ano-genital olfaction or stimulation, whereas females are motivated to produce USVs when males are further away, potentially as a mechanism to induce approach.

### DISCUSSION

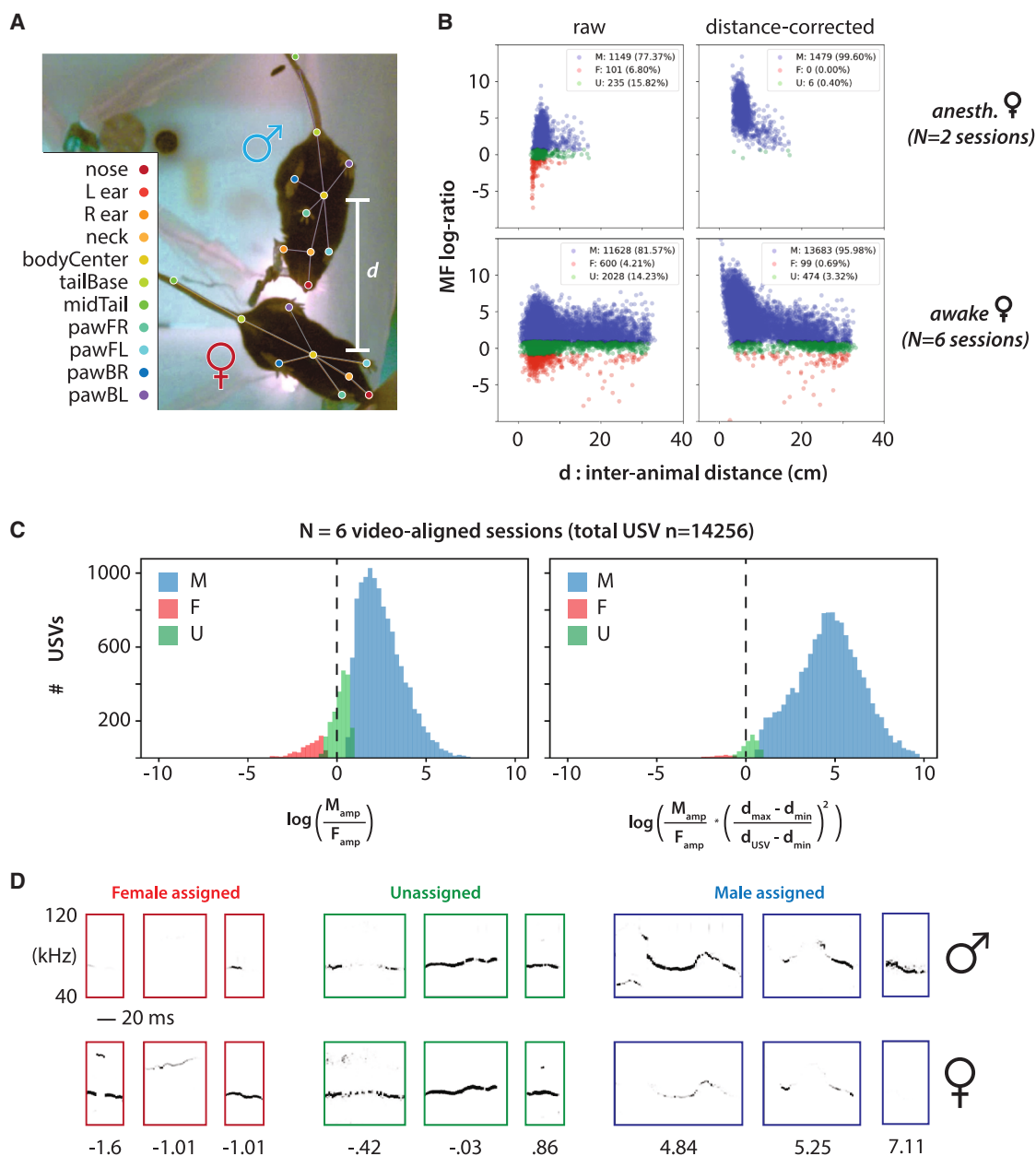
In this study, we present the first mountable ultrasonic microphones for use in studying mouse USVs. Wearable or mountable microphones have aided work in birds<sup>48</sup> and bats,<sup>49</sup> but adapting this technology for rodents has posed the challenge of finding an ultrasonic-capable microphone small enough to be comfortably carried by animals as small as laboratory mice. By identifying these lightweight microphones capable of recording USVs, developing custom attachment hardware and circuitry, and adding video tracking, we were able to assign a high percentage (90%–97%) of vocalizations to an individual mouse. Our findings serve as a proof of concept for the use of miniature microphones to study mouse social behavior, and we envision this system to be applicable to many different species.

Another group simultaneously developed the use of these miniature microphones in rats. However, there are important differences. They surgically implanted the microphone directly into the nostril and also had a head mount, which is a more invasive surgery and too large for mice.<sup>52</sup> In contrast, our system requires a minimally invasive attachment glued to the skull, which provides more surface area across multiple species for attachment of the microphone. Moreover, the dimensions of both the mounted headgear and microphone circuit board can be readily customized to accommodate any gross anatomical differences in other USV-producing species. Their method attached the miniature microphone to the nostril of one rat of a pair, where the other rat's vocalizations were picked up by an overhead large microphone and assigned based on subtraction of the recordings from the miniature microphone on the one animal; our method has miniature microphones attached to each animal. Finally, their method and other recent alternatives using microphone arrays rely on deep neural network models paired with manual correction to assign vocalization identity; our method uses a computationally simple comparison of amplitudes between a pair of mini-microphones. Therefore, while the recent study<sup>52</sup> validates the use of these microphones in measuring rodent USVs, our system provides a flexible, detachable microphone that is generalizable to multiple species, including mice, with a computationally simple and easily adoptable identity assignment method.

The major benefits of our system are that these mini-microphones are extremely low cost (\$0.60 each at the time of writing), easy to implement—requiring very little material and computational investment—and very customizable. We provide our designs of the circuit board and arm attachment, along with the materials list and procedures used. The utility of these mini-microphones also lies in their ease of integration with other systems. We demonstrated ready integration and enhanced USV assignment with video data. Our system has the potential to pair with an online USV assignment system, behavioral perturbations, electrophysiology, or optogenetics, which would allow

(E) Histogram showing male:female amplitude ratios zeroed to the respective gain-matched mean ratio for 22,561 USVs across 9 sessions. The majority of syllables are louder on the male mini-microphone (rightward skew to the red line).

(F) Examples of sonograms from female-assigned (red), unassigned (green), and male-assigned USV syllables and amplitude differences between mini-microphones of a male and female pair. Top row, recording from male mini-microphone; bottom row, recording from female mini-microphone. Black numbers show male:female amplitude ratios, and Roman numerals show position on amplitude ratio distribution in (E).

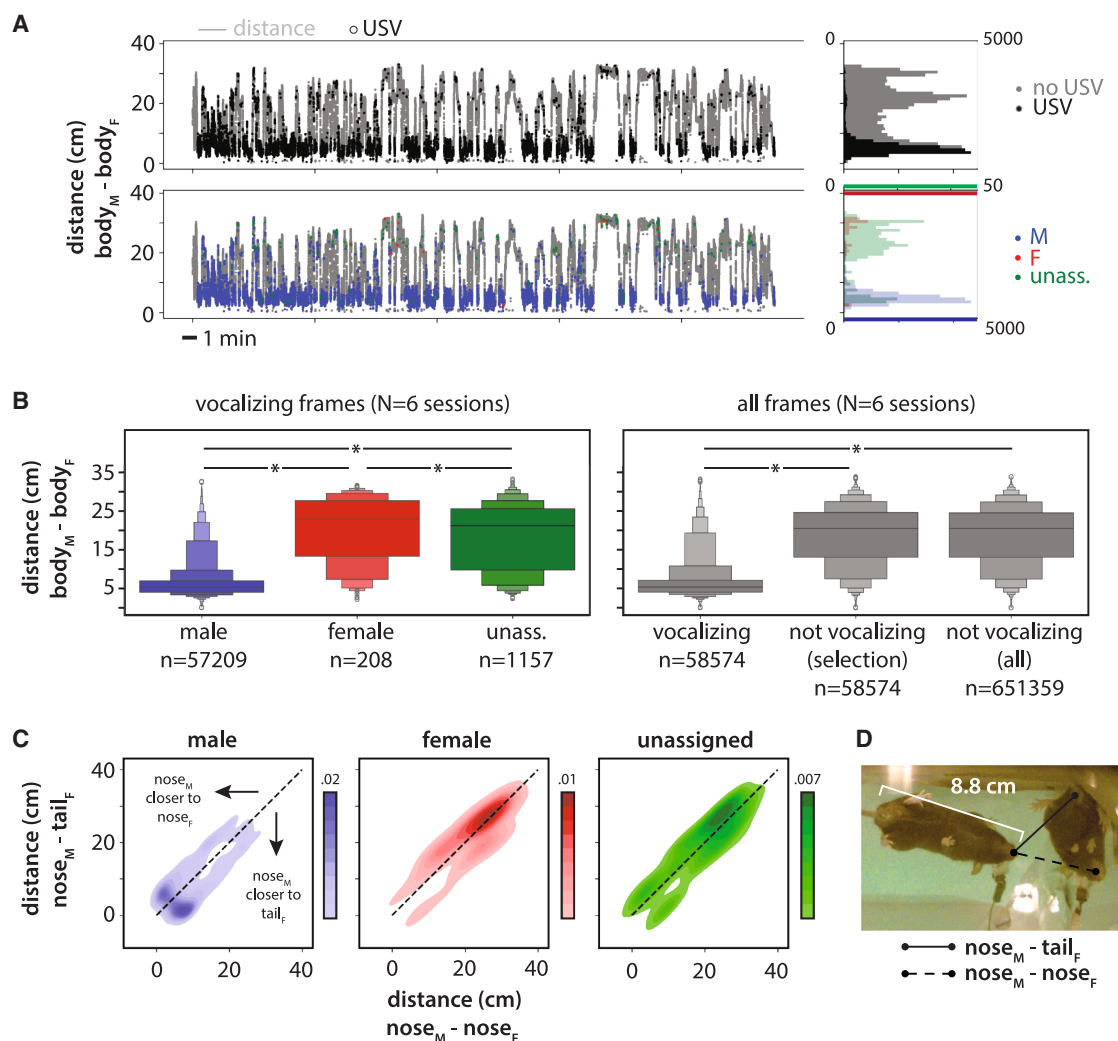


**Figure 3. Video tracking to refine vocalization assignment**

(A) Tracking mouse body parts during video recording using SLEAP. A total of 11 separate body features were defined and manually labeled.  
(B) M/F amplitude ratio vs. inter-animal distance. Color, USV assignments (male: blue, female: red, unassigned: green). Top, 2 control anesthetized female sessions. Bottom, 6 awake female sessions with concurrent video recording. Left, audio-alone assignment. Right, distance-corrected assignment.  
(C) Left, histogram of initial amplitude ratio distribution for 6 of 9 original sessions with aligned video. Right, amplitude ratios after distance correction for the same 6 sessions.  
(D) Example USVs from distance-corrected distributions. Top spectrograms, trace from male microphone. Bottom spectrograms, trace from female microphone. Bottom text, distance-corrected amplitude ratios.

in-depth investigation of multi-animal vocal behavior. While our method allows easy and high-fidelity USV assignment, existing array-based methods still have the benefit of featuring fully wireless behavior, which can accommodate animal recordings beyond a pair with greater ease. However, future versions of our system could involve a wireless setup with appropriate

high-speed data streaming or data logging. For an even less invasive approach, one such future system could make use of a necklace that is glued or tightened around the mouse's neck, thus not requiring surgery. Our microphone setup serves as an intrinsically modular system to study ultrasonic social behavior across many different species. The associated computationally



**Figure 4. Integration of video tracking with vocalization assignment**

(A) Inter-animal distance (gray) across a single recording, measured male body-centroid  $M$  to female body-centroid  $F$ . Top, video frames in which any USV was detected are labeled with black dots. Bottom, video frames in which male (blue), female (red), or unassigned (green) syllables are labeled with colored dots. Right of each distance plot, histogram of distribution of distances during occurrence of each type of vocal behavior.

(B) Comparing inter-animal distance across vocalization types. Left, plots showing distribution of body distance between animals during frames containing male (M), female (F), or unassigned (UN) vocalizations (independent t test; M vs. F  $p = 0$ , M vs. UN  $p = 0$ , F vs. UN  $p = 0.023$ ). Right, plot of all video frames showing distributions of distance for all vocalizations, an equivalent number of randomly selected non-vocalizing periods, and all non-vocalizing periods (independent t test; vocalizing vs. random nonvocalizing frames  $p = 0$ , vocalizing vs. all nonvocalizing frames  $p = 0$ , random nonvocalizing vs. all nonvocalizing frames  $p = 0.3$ ). Data shown aggregate six 20- to 40-min recordings with both audio and video data.

(C) Density plots comparing nose<sub>M</sub>-nose<sub>F</sub> and nose<sub>M</sub>-tail<sub>F</sub> distances for each close ( $\leq 8$  cm) USV. Unity line, dashed black line.

(D) Example of nose<sub>M</sub>-nose<sub>F</sub> and nose<sub>M</sub>-tail<sub>F</sub> distances.

and analytically simple method of identity assignment makes our system further adaptable to other systems. We believe that our method represents the ideal equipoise between acoustic fidelity, analytical complexity, and inherent modularity within existing methods of study for pairs of animals.

Our findings are consistent with some found in array microphone systems,<sup>42–45</sup> namely that male-assigned vocalizations often occur at very close range. Our finding that the majority of male USVs occur during ano-genital proximity (presumably with sniffing and stimulation) is consistent with the idea that

the male's USVs directed to a female is a courtship-facilitation behavior. With only paired mini-microphones without distance correction, our total USV assignment percentages (89.1%), as well our proportion of female-assigned vocalizations (5.6%), are in line with those of the highest-fidelity microphone array system (91.1% and 4.4%, respectively<sup>43</sup>). However, with our distance correction, our method assigns only 0.69% of USVs as female. Previous methods assigned female vocalizations at very close inter-animal distances, particularly during face sniffing<sup>43</sup>; thus, the relatively higher proportion of female USVs may be an

artifact and, in reality, produced by the male. Our low proportion after distance correction of female USVs is also in line with a recent study that found that USV rates from females paired with muted males were extremely low.<sup>53</sup> It is also possible, due to the similarity in behavioral profiles between the unassigned and female-assigned USVs, that many of the unassigned USVs are produced by the female (Figure 4C).

We were able to perform USV assignment that rivals more resource- and computationally intensive methods with only two microphones and simpler computations. However, further innovations with this system might enable an even higher rate of USV assignment. These future directions might include placing a mini-microphone on both sides of an individual mouse's head; providing a second equidistant microphone would likely aid in assigning vocalizations produced at close snout-to-snout proximity but at an angle, potentially eliminating the need for video tracking. A non-wired or non-surgical harness would also allow more naturalistic conditions for the vocal and social behavior and would permit even longer recording sessions without complications from cable tangling. The video recordings may not only be useful for investigating behavioral components that are coordinated with mouse vocal production behavior but could also serve as an additional source of triangulation information. Finally, we intentionally kept the identity-separation computations as simple as possible. We encourage other groups to experiment with variations on the system that may further improve its USV assignment capabilities.

### Limitations of the study

This method is best suited for pairs of vocalizing animals and involves wired microphones. For larger groups, absent a more complex commutator system to avoid wire tangling, overhead array-based methods are more suitable. Similarly, if the USV behavior being studied requires completely wire-free behavior, an array-based method or an additional data logger would be necessary. While this method does not impede on the function of the brain, ears, or nostrils of mice, the surgical implantation is still invasive to the skin and skull surface. To improve accuracy even further without a video system, one could employ systems used to cancel background vocalizations coming from others in close proximity. We also did not record video for all sessions, so 3 of the original 9 sessions were not distance corrected. Further, because we started video recording and alignment separately from the audio, some USVs within the 6 sessions for which we do have video were not able to be corrected. Our animal assignment validation test used anesthetized females, which cannot produce USVs but also cannot move. Although the males could and do still move around the cage when vocalizing in the presence of an anesthetized, non-vocalizing female, it is possible that some aspects of the naturalistic vocal behavior interaction in pairs of mice will be different, as in prior studies from our group, males produce a higher proportion of complex multiple jump syllables in the presence of awake vs. anesthetized females.<sup>24</sup> However, singing a differential proportion of syllable types in the presence of an awake, devocalized female is unlikely to change any incorrect animal assignment results. We attempted surgical devocalization by recurrent laryngeal nerve (RLN) transection,<sup>54,55</sup> but as found in past studies in mice, unilateral

RLN resulted in the animals still being able to vocalize, and bilateral RLN transection caused immediate asphyxiation.<sup>56,57</sup> While our study was submitted, another group reported using transgenic lines to perform activity-dependent ablation of USV-controlling midbrain neurons,<sup>53</sup> and the results of devocalized moving females led to the same conclusion as our findings with anesthetized non-vocalizing females. Finally, while mini-microphones can be re-used many times, after recording sessions with high rates of unassigned USVs, it may be necessary to replace microphones to continue to ensure higher rates of assignment.

### RESOURCE AVAILABILITY

#### Lead contact

Further information and request for resources should be directed to the lead contact, Elena N. Waidmann ([ewaidmann@rockefeller.edu](mailto:ewaidmann@rockefeller.edu)).

#### Materials availability

The miniature microphone system is composed of components listed in the [key resources table](#). Instructions on assembly can be found in the [method details](#). Circuit board and arm attachment files can be found in Dryad Database DOI: [10.5061/dryad.kpr4xhhh](https://doi.org/10.5061/dryad.kpr4xhhh).

#### Data and code availability

- Audio and video data have been deposited at Dryad and are publicly available as of the date of publication. Dryad Database DOI: [10.5061/dryad.kpr4xhhh](https://doi.org/10.5061/dryad.kpr4xhhh).
- All original code has been deposited at Dryad and is publicly available as of the date of publication. Dryad Database DOI: [10.5061/dryad.kpr4xhhh](https://doi.org/10.5061/dryad.kpr4xhhh).
- Any additional information required to reanalyze the data reported in this paper is available from the lead contact upon request.

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### AUTHOR CONTRIBUTIONS

E.N.W. and E.D.J. conceived of the experiment. E.N.W. and V.H.Y.Y. developed custom hardware. E.N.W., V.H.Y.Y., E.L., and W.C.D. performed surgeries, collected behavioral data, and performed data analysis. E.N.W., V.H.Y.Y., and E.D.J. wrote the manuscript.

### DECLARATION OF INTERESTS

The authors declare no competing interests.

### STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

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- Mini-microphone assembly
- Mini-microphone attachment surgery
- Behavioral paradigm
- **QUANTIFICATION AND STATISTICAL ANALYSIS**
  - Audio/video alignment
  - Video tracking and inter-animal distance measurements
  - Analysis of identity assignment

## SUPPLEMENTAL INFORMATION

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## STAR★METHODS

### KEY RESOURCES TABLE

| REAGENT or RESOURCE                           | SOURCE                 | IDENTIFIER                                                                                                                               |
|-----------------------------------------------|------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Experimental models: Organisms/strains</b> |                        |                                                                                                                                          |
| Mouse: C57BL/6J                               | The Jackson Laboratory | JAX #000664                                                                                                                              |
| <b>Software and algorithms</b>                |                        |                                                                                                                                          |
| Bonsai                                        | Bonsai Foundation      | RRID: SCR_017218                                                                                                                         |
| Avisoft-RECORDER                              | Avisoft Bioacoustics   | RRID: SCR_014436                                                                                                                         |
| Avisoft-SASLab Pro                            | Avisoft Bioacoustics   | RRID: SCR_014438                                                                                                                         |
| Mouse Song Analyzer 2.0                       | Stoumpou et al. (47)   | <a href="https://github.com/peterfschade/MSA2">github.com/peterfschade/MSA2</a>                                                          |
| <b>Other</b>                                  |                        |                                                                                                                                          |
| Printed Circuit Board: Polyimide Flex 1 layer | PCBWay                 | (see KiCad file in SourceMaterials.zip, Dryad Dataset DOI: <a href="https://doi.org/10.5061/dryad.kpr4xhhh">10.5061/dryad.kpr4xhhh</a> ) |
| Knowles MEMs Microphone                       | Mouser Electronics     | SPU0410LR5H-QB                                                                                                                           |
| Ultrasound Gate 416H recording interface      | Avisoft Bioacoustics   | Part # 34163, 34164                                                                                                                      |
| Ultrasonic Dynamic Speaker (Vifa)             | Avisoft Bioacoustics   | Part # 60108                                                                                                                             |
| 216H Ultrasound Playback Interface            | Avisoft Bioacoustics   | Part # 70117, 70118                                                                                                                      |
| Basler Ace 2 Camera                           | Basler AG              | a2A1920-160ucPRO                                                                                                                         |
| <b>Deposited data</b>                         |                        |                                                                                                                                          |
| All audio/video data, code, and schematics    | Dryad                  | Dataset DOI: <a href="https://doi.org/10.5061/dryad.kpr4xhhh">10.5061/dryad.kpr4xhhh</a>                                                 |

### EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS

#### Mouse lines

Adult male and female C57BL/6J mice were purchased from Jackson Laboratories (JAX #000664), and were 3–5 months of age at the time of recording. Prior to the surgical procedure, mice were group-housed in 5 animals per cage of the same sex, then individually housed after the procedure and between recording periods. The mice had free access to food and water. All animal treatment adhered to standards laid out by National Institute of Health guidelines, with approval from The Rockefeller University Institutional Animal Care and Use Committee (IACUC).

### METHOD DETAILS

#### Mini-microphone assembly

We applied solder paste to the pads of each printed circuit board (PCB) with a custom stencil (PCBWay). We placed a JST connector, 0.1μF capacitor, and mini-microphone (Knowles SPU0410LR5H-QB) on their corresponding pads and ran through a reflow oven to affix the parts. We confirmed appropriate electrical connectivity of each mini-microphone PCB with a multimeter. We then glued a custom 3D-printed arm to the back of the PCB (For 3D-print and circuit files, see Supplementary materials).

#### Mini-microphone attachment surgery

To securely attach the custom mini-microphone, we used a locknut and screw design to allow convenient attachment and removal of the mini-microphone as needed between recording sessions. A steel serrated flange locknut (Zinc-Plated, 4–40 Thread, 1/4" Wide from McMaster-Carr) was placed on the head of the mice in a minimally invasive surgical procedure. Aseptic techniques were observed throughout the surgery, and surgical tools and the locknut to be implanted were steam sterilized using a 34-min autoclave gravity cycle at 121°C.

For the surgery, anesthesia was induced using isoflurane and the mouse was placed in a stereotaxic frame, secured with a nose cone and ear bars. Ear bars make the surgery easier and faster to perform, but are not strictly necessary, as no stereotaxic coordinates were used, and it is possible to place the animal in only the nose cone. We used ear bars for the initial surgeries, and for additional controls (Table S1) were performed surgeries without ear bars. A heating pad was placed under the mouse body to maintain appropriate body temperature throughout the procedure, and an ophthalmic ointment was applied to the eyes. Hair on the top of the head was trimmed and removed using a small amount of hair removal cream. The exposed skin was cleaned with an iodine swab stick then a sterile alcohol prep pad, repeated three times. An incision was made on the skin above the skull, creating an opening

running anterior-posterior over the skull, roughly between the frontonasal suture on the anterior portion and lambdoid suture on the posterior portion. A hemicycle cut was made along either side of the opening to create an oval skin opening centered approximately on bregma. The membranes on the skull were removed using a combination of scraping with fine surgical forceps and application of a small amount of 3% hydrogen peroxide. As an analgesic to help with postsurgical recovery, a small amount of 0.25% Bupivacaine (2.5 mg/mL) was applied to the skull and edges of the skin opening using a syringe. Once the opening was cleaned with bacteriostatic 0.9% sodium chloride saline and dried, 3M Vetbond was used as a biocompatible adhesive to seal the skin opening to the skull.

After the Vetbond dried, the locknut was secured to the surface of the skull using a small amount of Loctite Superglue, centered on the oval skin opening. To further stabilize the locknut and cover all exposed skin, we used OrthoJet dental cement to cover the flange surrounding the locknut and all skin edges. Three layers of OrthoJet were applied, ensuring the next layer is applied only when the preceding one had dried completely. The mouse was then placed in a clean cage on a heating pad for recovery, provided with hydrogel and food, and closely monitored in the next days for potential complications. The procedure used was identical between male and female mice.

### Behavioral paradigm

Recording sessions were conducted with a male and female mouse together, after both had received the locknut implant described above and recovered appropriately. During a recording session, we connected 3 separate channels to the Ultrasound Gate 416H (USGH) recording interface: the male mini-microphone, the female mini-microphone, and an overhead microphone (Avisoft Bioacoustics CM24/CMPA) suspended on the top of the acrylic box, 15 inches from the floor of the box. Mini-microphones were connected to the USGH via a 5-pin XLR to screw terminal adapter (Sescom). We used the custom skull attachment to secure the custom flexible printed circuit board (PCB) (PCBWay) with the Knowles MEMS microphone (see Supplementary material for circuit design). The 15 inches from the bottom ensures that the overhead microphone captures the full spatial coverage of the base surface area (12 × 12 inches) of the box. All audio recordings were conducted using the Avisoft-RECORDER software (Avisoft Bioacoustics) on a Windows PC. Video data was recorded from below using the Basler Ace 2 camera (a2A1920-160ucPRO), and from above with a Firefly S camera (Teledyne FLIR). Bonsai was used to record and process videos. Video and audio recordings received simultaneous 1 Hz pulses to enable post-hoc synchronization.

Prior to the mouse social USV recording, the 2 mini-microphones were gain-matched to ensure a comparable level of sensitivity to the audio being received. The two mini-microphones were attached to a wall of the recording box next to each other, equidistant to an Ultrasonic Dynamic Speaker (Vifa #60108, Avisoft Bioacoustics) connected to a 216H Ultrasound Playback Interface. An audio file of sample ultrasonic vocalizations was played, while the Avisoft-RECORDER spectrograms for the two mini-microphones were visually monitored and the gain for each respective spectrogram channel was adjusted to a close match. A short recording from the mini-microphones was then recorded so it could be used as a reference point during data analysis.

To secure the PCB with the mini-microphone to the head mount on the mice, the mice were first briefly anesthetized using isoflurane. The PCB, glued to a custom 3D printed acrylic mount was placed onto the locknut on the mouse's head, and secured using a 4–40 Phillips head screw (McMaster Carr; See Supplementary Materials for 3D designs). The PCB and mini-microphone were oriented with the acoustic port facing directly toward the mouse's mouth (0° angle, see [Figures 2G](#) and [S1H](#)). The mouse was then moved to the recording setup and the mini-microphone wire was attached to the sound channel; the mouse was given around 10 min to recover from anesthesia and acclimatize to the environment. The procedure was identical for both male and female mice. The audio and video recordings were started once the female's microphone wire was connected and she was placed inside the recording box, after having connected the male mouse. A 40-min recording was conducted in this social context, with constant monitoring to ensure the cables were not tangled as to impede the free movement of either mouse in the box. If cables began to tangle, we manually rotated a small 3D printed piece ([Figure 2A](#)) through which both mini-microphone cables pass in separate channels; this rotation allowed any tangling to transfer elsewhere on the wires. So far this has been a workable solution for pairs of 2 mice even over an hour of recording. Once the recording session was complete, the mice were unplugged from their cables, and the screw loosened to remove the acrylic mount/PCB attachment. The mice were then returned to their individual housing cages until the next session.

## QUANTIFICATION AND STATISTICAL ANALYSIS

### Audio/video alignment

An AdaFruit Feather M0 Basic was programmed to generate 1Hz digital pulses, with a length randomly selected from 100ms, 200ms, 300ms, 400ms, and 500ms. For videos, the digital pulses were delivered to two LEDs that were positioned within clear view of the top and bottom camera, respectively. For audio, the digital pulses were delivered to the digital input port on the Avisoft USGH 4-channel recorder used to collect recording from all microphones. For each recording modality, frames containing pulse starts and ends were identified. We confirmed the pulse patterns matched, then created a global time frame that progressed by 1 s with each pulse start. For each 1-s block, for each modality, the number of F frames was counted, and individual frames were assigned a time increment 1/F. Thus audio and video were both aligned to the same uniform external source and every frame was placed within a shared global time frame.

### Video tracking and inter-animal distance measurements

We trained a multi-animal bottom-up SLEAP model<sup>51</sup> on 527 total frames across seven videos taken from the bottom view camera (see Figures 3A and S4A). We labeled 11 parts (nose, neck, left ear, right ear, body center, tail base, middle tail, right forepaw, left forepaw, right rear paw, left rear paw) but for this study we only used the nose, neck, body center, and tail base. After training was completed, we ran model inference for each video, and manually corrected any identity swaps between male and female skeletons. We measured the left-to-right and front-to-back lengths inside of the recording arena in centimeters, and in pixels with the first frame of each video. We used these pixel/centimeter ratios to convert animal distances output from SLEAP in pixels to centimeters.

### Analysis of identity assignment

For each recording session, we generated spectrograms ( $f_s = 250\text{e}3$ , Hann window,  $n_{\text{perseg}} = 256$ ) from the wav files for each channel (top, male mini, female mini). For all recordings, we used the spectrogram output and applied no additional filtering. We used the top recordings as a ground truth for acoustic and temporal properties, and the mini-microphone recordings to compare volume differences for each individual syllable to use for identity assignment. We used MSA 2.0<sup>50</sup> to perform general acoustic and temporal analysis of the top microphone recordings for both the gain-matching session and courtship session. From MSA 2.0, we obtained the start and end times of each USV syllable on the top recording. Within each syllable, for each time bin, we identified the three frequencies that had the highest amplitude in the top microphone recorded spectrogram (Figure 2B). For that same time bin, we measured the mean amplitude across those three frequencies, in the male and female mini-microphone spectrograms. We calculated the median amplitude across all time bins for a single syllable, for male ( $M_{\text{amp}}$ ) and female ( $F_{\text{amp}}$ ) spectrograms.

Once we obtained male and female amplitudes for every USV, we took the log of the ratio between male and female:  $\log(M_{\text{amp}}/F_{\text{amp}})$  for each syllable. Here, an amplitude ratio greater than 0 means that a single USV was louder on the male mini-microphone, while a amplitude ratio below 0 indicates that the USV was louder on the female mini-microphone. A amplitude ratio value of 0 signifies no volume differences.

The  $\log(M_{\text{amp}}/F_{\text{amp}})$  amplitude ratio was computed, as above, for every syllable from the gain-matching session. We computed the mean ( $\text{gain}_{\text{mean}}$ ) and standard deviation ( $\text{gain}_{\text{SD}}$ ) across all syllables for a single gain-matching session. These values were used as a ground truth measurement of inherent amplitude differences between the two mini-microphones for each session. Because every courtship recording session was preceded by a gain-matching session, we used the same-day session as the amplitude ratio benchmark. We classified a single courtship USV<sub>x</sub>, that had amplitude-ratio  $\log(M_{\text{amp}}/F_{\text{amp}})_x$  according to the following logic:

$$\text{Male USV } \log(M_{\text{amp}}/F_{\text{amp}})_x \geq \text{gain}_{\text{mean}} + \text{gain}_{\text{SD}}$$

$$\text{Female USV } \log(M_{\text{amp}}/F_{\text{amp}})_x \leq \text{gain}_{\text{mean}} - \text{gain}_{\text{SD}}$$

$$\text{Unassigned USV } \text{gain}_{\text{mean}} - \text{gain}_{\text{SD}} \leq \log(M_{\text{amp}}/F_{\text{amp}})_x \leq \text{gain}_{\text{mean}} + \text{gain}_{\text{SD}}$$

For our distance-correction metric, the inter-animal distance during each USV ( $d_{\text{USV}}$ ) was computed between animal centroids, then normalized to maximum and minimum distance ( $d_{\text{max}}$ ,  $d_{\text{min}}$ ). The distance-correction scalar was applied to our amplitude ratio, as follows:

$$\log\left(\left(M_{\text{amp}}/F_{\text{amp}}\right) * \left(d_{\text{max}} - d_{\text{min}}/d_{\text{USV}} - d_{\text{min}}\right)^2\right)$$